



Sage: Smart Academic Hiring System

Literature Review Report

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The Department of Computer Science, National University of Computer and Emerging Sciences, accepts this thesis titled “SAGE : Smart Academic Hiring System”, submitted by Muhammad Abdullah (P22-9321), Ashar Naveed (P22-9346), and Dilawar Khan (P22-9099), in its current form, and it satisfies the dissertation requirements for the award of the Bachelor of Science Degree in Computer Science.

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Muhammad Abdullah

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Abstract

This document presents the literature review for the **Sage (Smart Academic Hiring System)** Final Year Project. Sage automates faculty hiring at FAST University using resume parsing, candidate-job matching via an improved Gale–Shapley algorithm, automated LLM-based testing, and AI interview evaluation focusing on non-technical attributes such as confidence and personality. This review examines related research on resume parsing, candidate-job matching, interview automation, and fairness in AI recruitment, highlighting existing gaps addressed by Sage.

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Chapter 1

Introduction

1.1 Background

Recruitment and selection of faculty is a time-consuming and often subjective process. Universities typically rely on manual screening of CVs, ad-hoc interviews, and inconsistent evaluation criteria. Recent advances in Natural Language Processing (NLP), Machine Learning (ML), and Large Language Models (LLMs) now enable automation of hiring tasks such as CV parsing, semantic matching between candidate profiles and job descriptions, automated testing, and even preliminary AI-conducted interviews.

1.2 Motivation

Sage automates the entire recruitment process for academic hiring at FAST University. By integrating structured CV parsing, a stable-matching algorithm (Gale–Shapley), LLM-powered test generation and grading, and an AI interviewer for behavioral assessment, Sage reduces HR workload, improves fairness, and ensures consistency in faculty selection.

1.3 Scope of the Literature Review

This literature review focuses on:

- Resume/CV parsing and NLP-based information extraction.
- Candidate–job matching via the Gale–Shapley stable matching algorithm.
- Automated testing using LLMs for question generation and grading.
- AI-driven interview evaluation.
- Fairness and bias mitigation in automated hiring.

Chapter 2

Review of Literature

2.1 Method for Selecting Papers

Peer-reviewed studies from IEEE, Springer, Elsevier, and arXiv (2020–2025) were reviewed. Selection criteria included technical relevance to AI recruitment, matching algorithms, and ethical automation.

2.2 Comparative Analysis of Related Work

Table 2.1 summarizes major studies relevant to Sage—covering Gale–Shapley algorithm variants, resume parsing systems, AI interview bots, and fairness in automated recruitment.

Year	Paper / System	Methodology	Dataset / Evaluation	Main Findings	Gaps / Limitations
2020 (Est.)	Performance-based Stable Matching using Gale-Shapley Algorithm	Application of the Gale-Shapley algorithm for employee placement using performance metrics and institutional preferences.	Conceptual study for employee position balancing.	Demonstrated stable matching reduces dissatisfaction from misplacement.	Focused on internal placement; does not address external hiring like Sage .
2021	Interview Bot for Improving Human Resource Management	AI/ML-driven chatbot automating interview conduction and evaluation.	Conceptual design and feasibility study.	Proved AI chatbots can reduce bias and improve interview scalability.	Lacked large-scale testing; text-based only, not multimodal like Sage .
2022	Enhancement of Gale-Shapley Algorithm with Imbalanced Sets	Modified Gale-Shapley for unbalanced sets (candidates vs jobs) using duplication for stable pairing.	Theoretical study.	Extended stable matching to practical hiring imbalance.	Relied on duplication; Sage instead uses weighted scoring to balance matches.
2022	People versus Machines: Introducing the HIRE Framework	Systematic review of 22 empirical AI recruitment studies proposing HIRE framework.	Literature synthesis.	Found AI comparable or superior to human recruiters in efficiency.	Identified mistrust and ethical concerns; conceptual framework only.
2024	Automated Resume Parsing and Ranking using NLP	Transformer (BERT)-based NLP model for resume parsing and ranking.	Internet resume datasets.	Improved accuracy for skill and education extraction.	General-purpose parser; not optimized for academic CVs as in Sage .
2024	Putting Gale & Shapley to Work: Guaranteeing Stability Through Learning	Learning-based stable matching ensuring stability under uncertain preferences.	Theoretical simulations.	Provided stability guarantees in uncertain preference settings.	Purely theoretical; lacks domain weighting like in Sage .
2025	Optimizing Recruitment with AI: A Smarter Approach to Candidate Evaluation	AI-driven recruitment pipeline automating screening and interviews.	System design prototype.	Increased efficiency and accuracy in recruitment.	General recruitment; lacks academic-domain adaptation of Sage .
2025	Fairness in AI-Driven Recruitment: Challenges, Metrics, Methods, and Future Directions	Systematic review of bias metrics and mitigation in AI hiring.	Review of 120+ studies.	Summarized fairness metrics and mitigation frameworks.	Conceptual; Sage incorporates fairness-aware scoring to address this.

Table 2.1: Comparative Literature Review for Sage – Smart Academic Hiring System.

Chapter 3

Project Vision

3.1 Problem Statement

Manual shortlisting and interview scheduling in academic hiring is slow and inconsistent. Candidate evaluation is often subjective and hard to audit. Sage addresses these issues by automating extraction, matching, testing, and preliminary interviewing while keeping HR oversight.

3.2 Business Opportunity

Automation reduces HR time, speeds up hiring cycles, and yields more consistent candidate assessments. For a large university with recurring hiring cycles, Sage can materially reduce administrative costs and improve candidate–role fit.

3.3 Objectives

1. Build a CV parsing module for academic resumes.
2. Implement a candidate-job matching ranking engine.
3. Integrate LLM-based test generation and auto-grading for objective/short-answer questions.
4. Develop an AI interviewer to evaluate non-technical attributes and generate recorded sessions.
5. Provide transparent dashboards for HR with audit trails and candidate explanations.

3.4 Project Scope

Sage focuses on initial shortlisting, automated testing, AI interviewing for non-technical attributes, and final rank-generation for HR review. It does not perform final hiring approvals or legally binding decisions.

3.5 Constraints

Data privacy regulations, limited access to diverse labeled interview datasets, variability in CV formats, and the need to avoid biased models are the main constraints.

3.6 Stakeholders

- HR Staff — primary users who control hiring workflows.
- Candidates — supply CVs and take tests/interviews.
- Department Heads — view ranked lists and interview recordings.
- University Administration — final approvals and policy oversight.

Chapter 4

Diagrams and Placeholders

The following pages are placeholders for diagrams that you will insert later. Replace the framed boxes with actual diagrams (PDF/PNG/SVG) and adjust captions.

4.1 Use Case Diagram

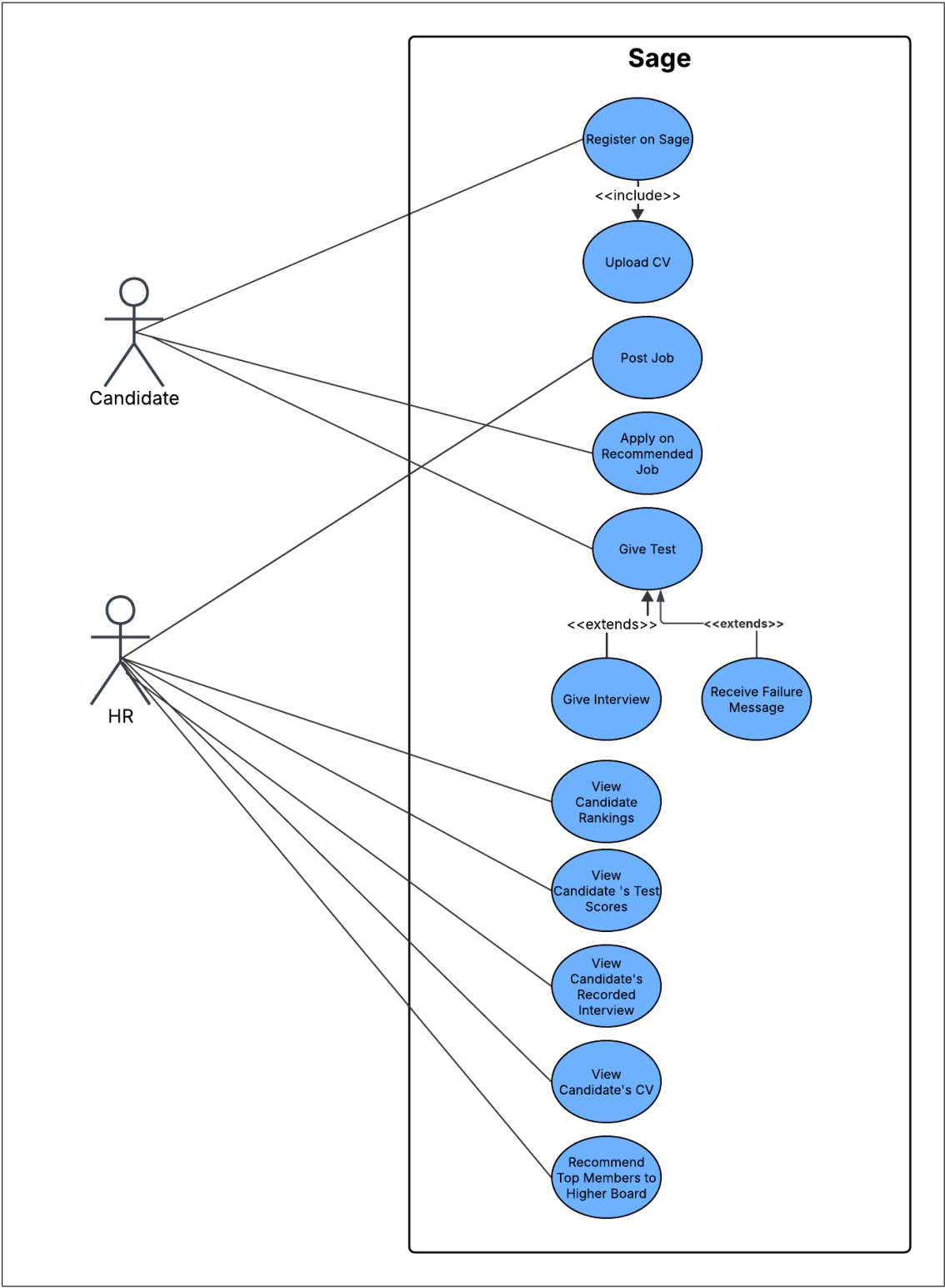


Figure 4.1: Use Case Diagram: High-Level System Interactions.

4.2 Sequence Diagram

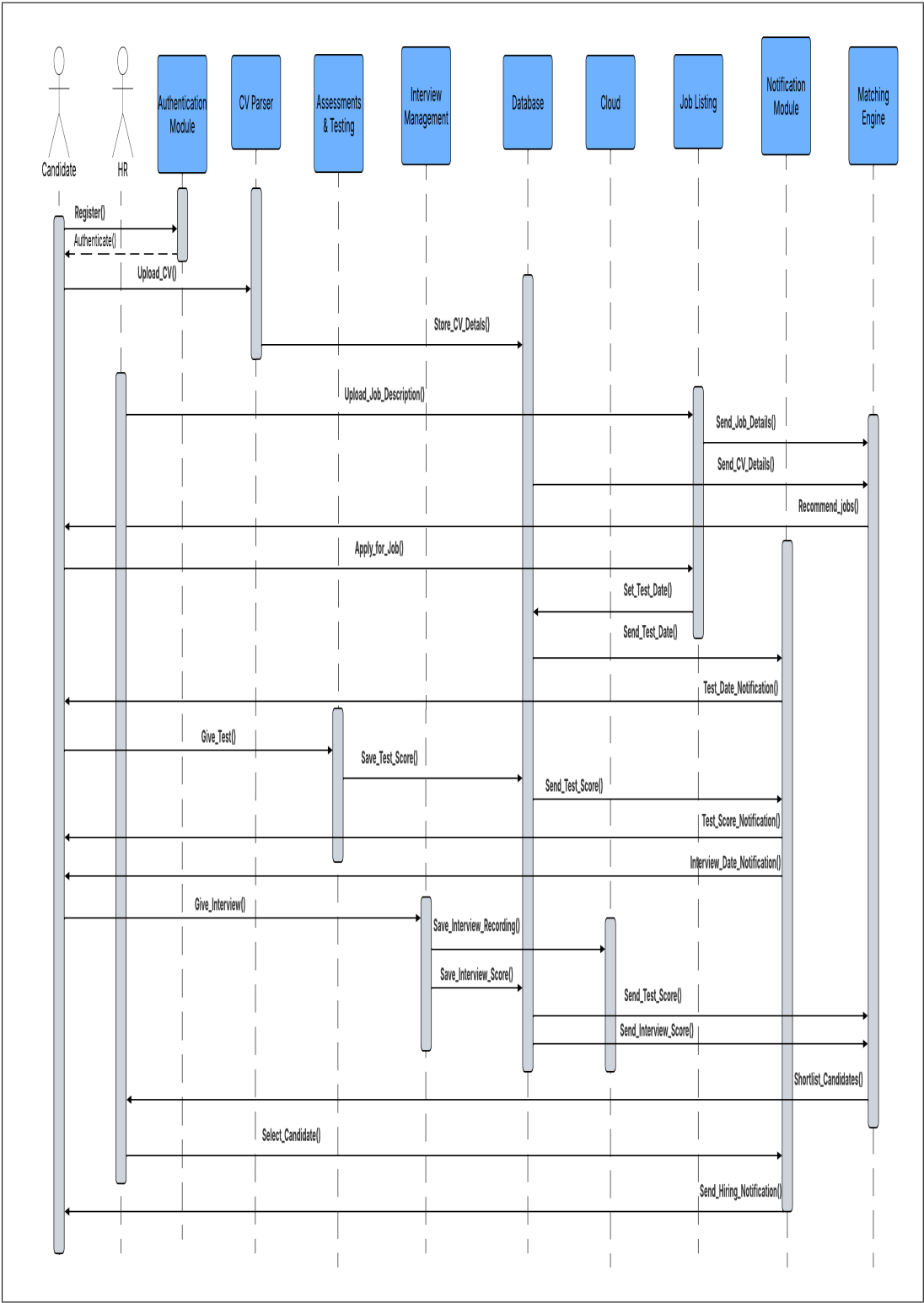


Figure 4.2: Sequence Diagram: Candidate Registration and Screening Process.

4.3 Data Flow Diagram (DFD)

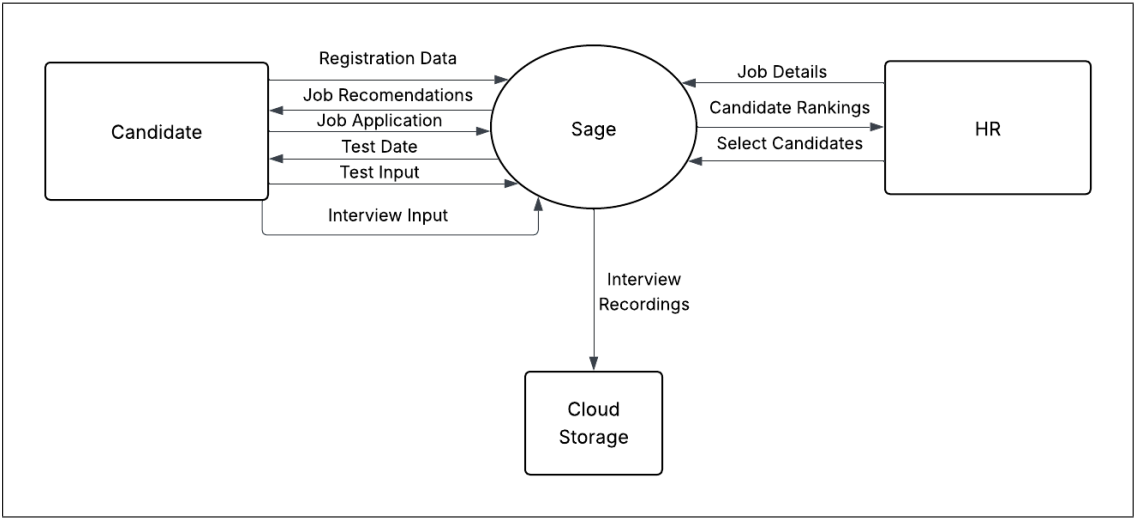


Figure 4.3: Data Flow Diagram — Level 1: System Data Flow Overview.

4.4 Activity Diagrams

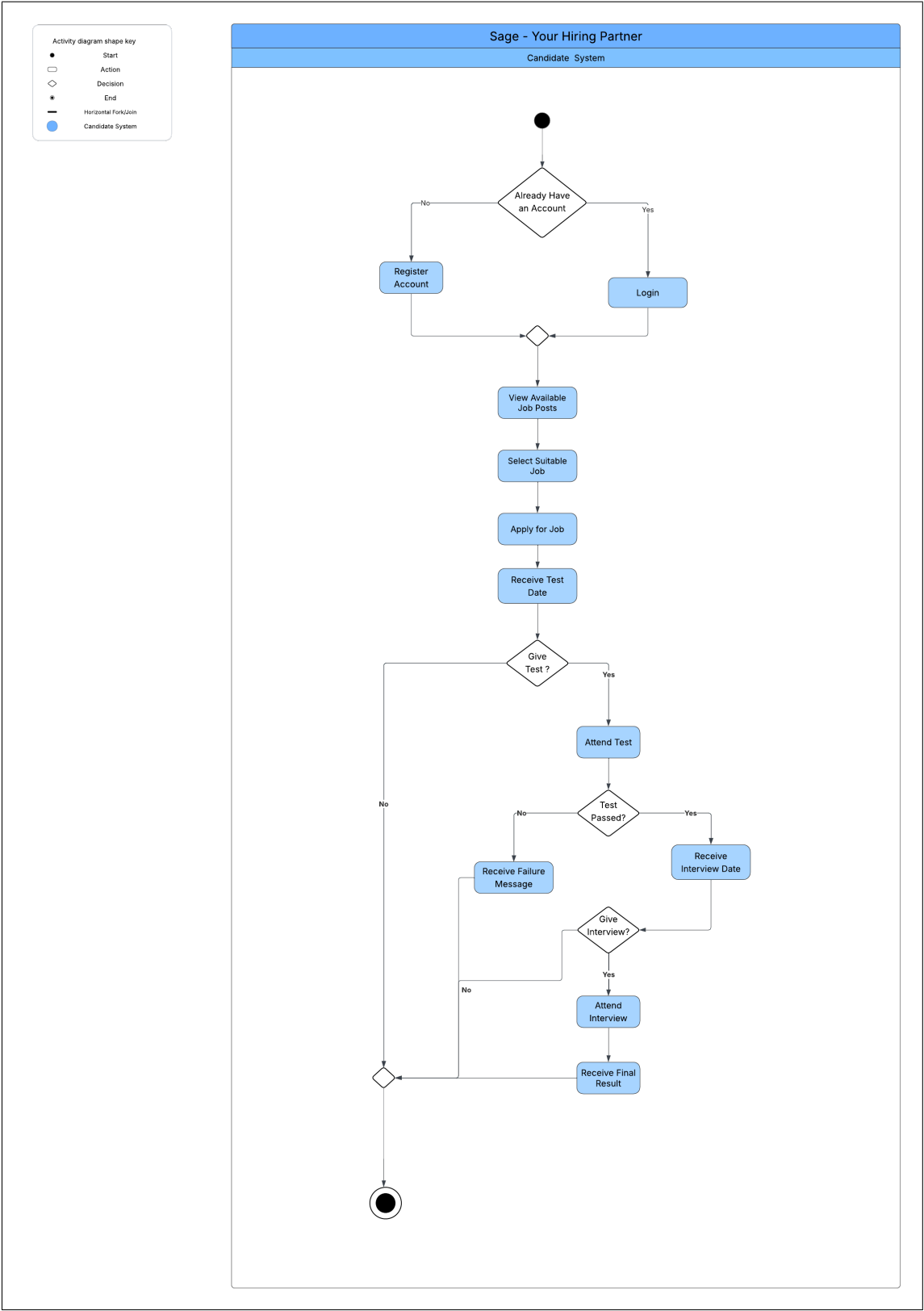


Figure 4.4: Activity Diagram: Candidate-System Interaction for Application.

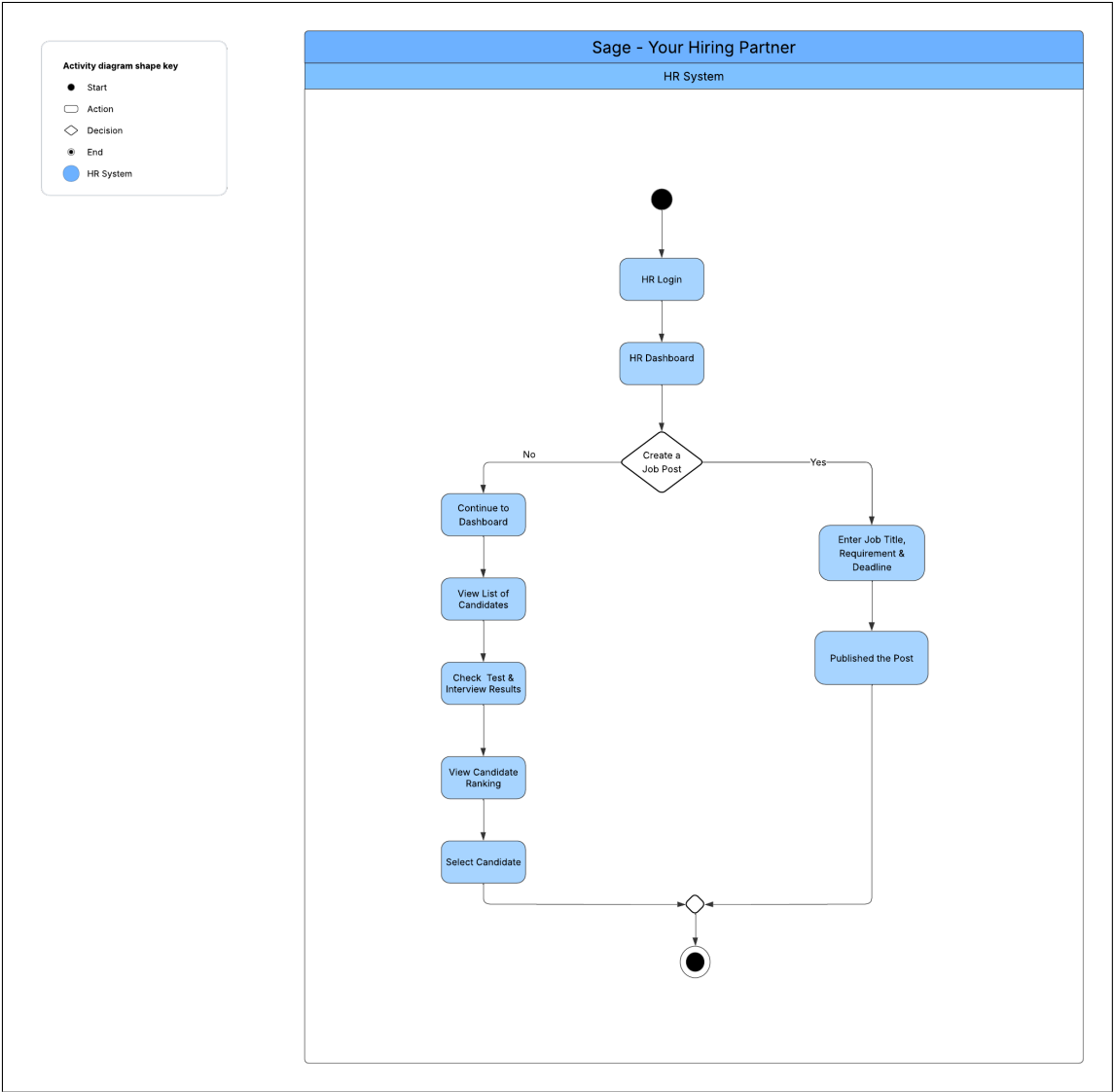


Figure 4.5: Activity Diagram: HR Staff Review and Final Matching Process.

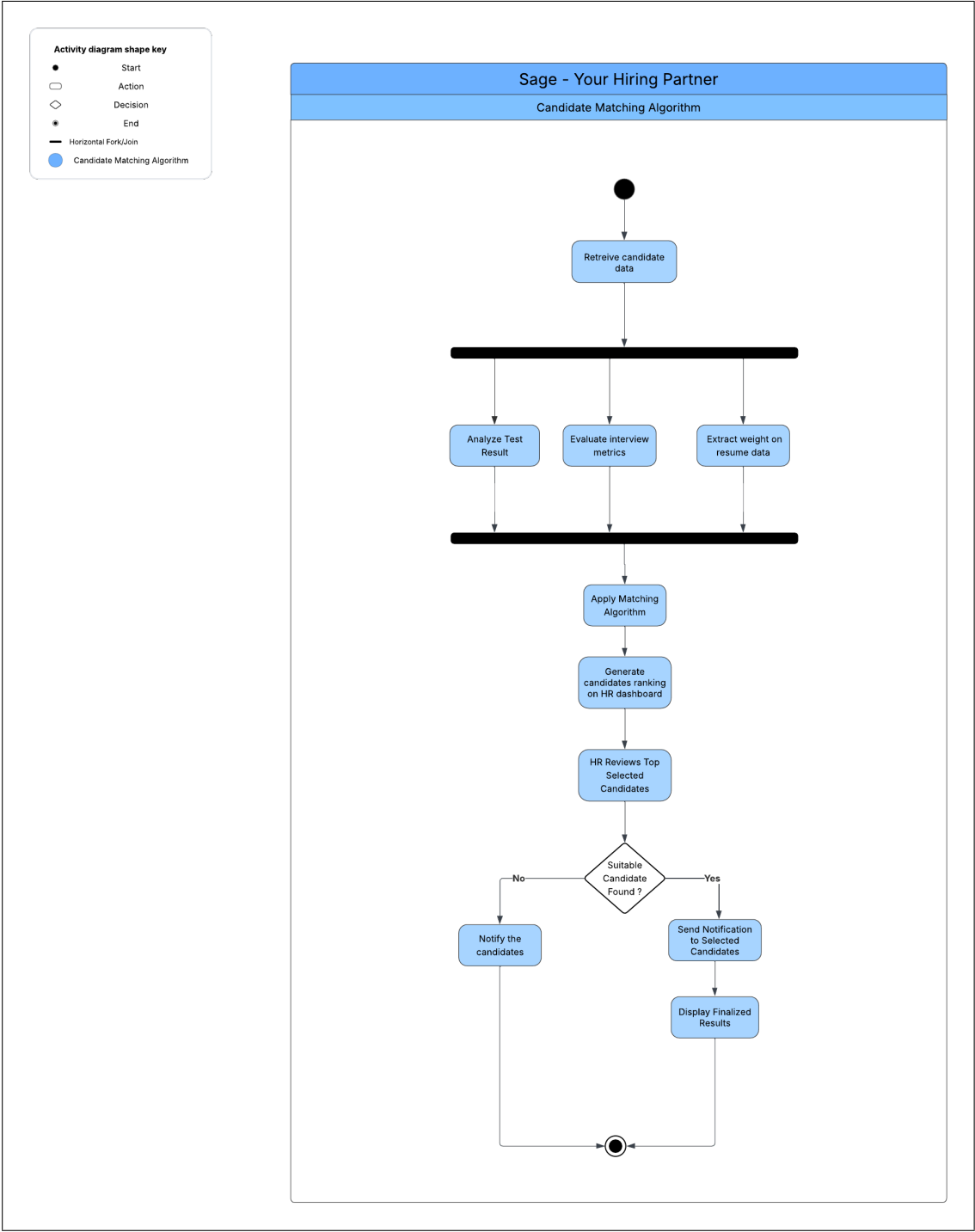


Figure 4.6: Activity Diagram: Candidate Interview Workflow.

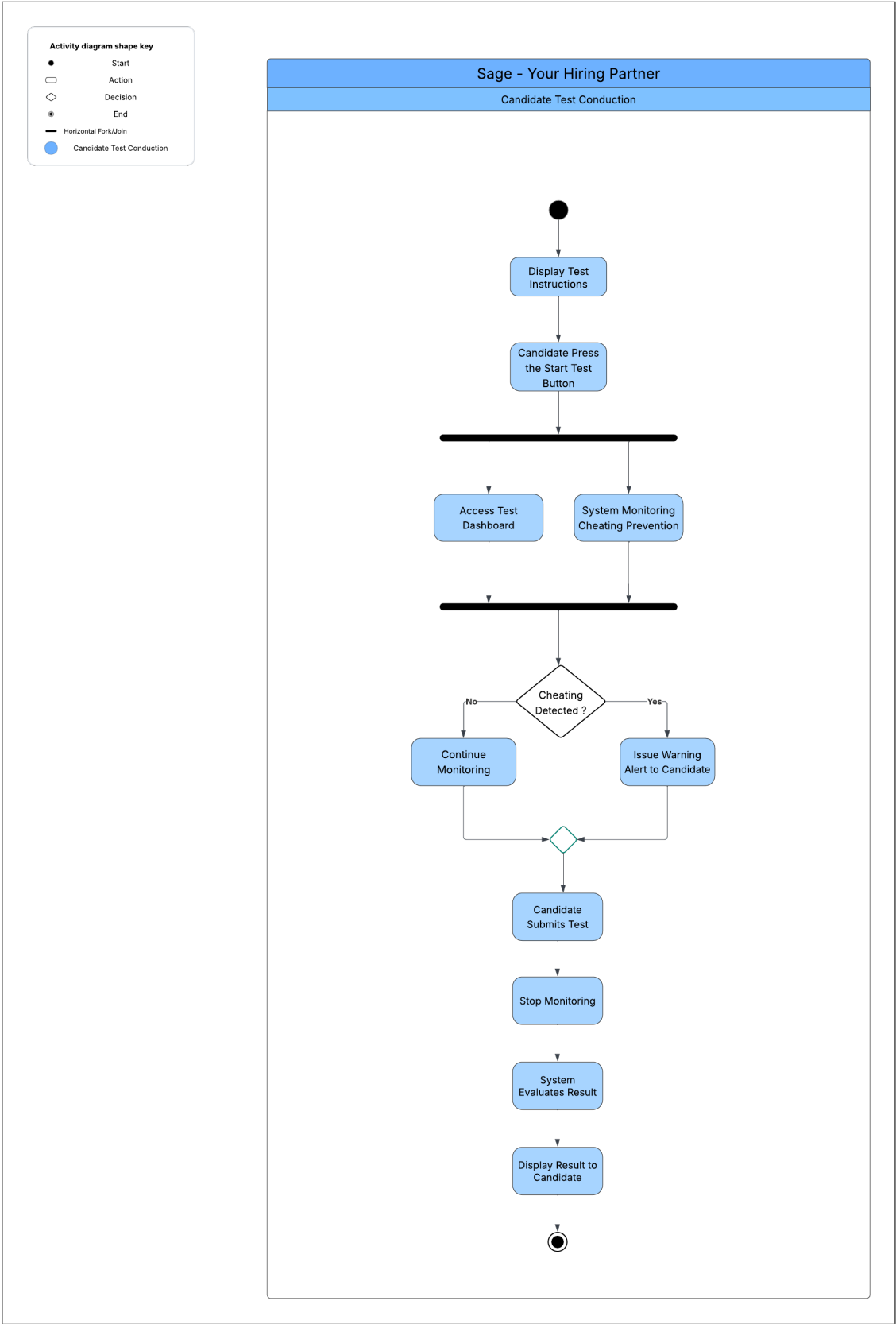


Figure 4.7: Activity Diagram: Candidate Test Generation and Submission.

4.5 Layer / Architecture Diagram

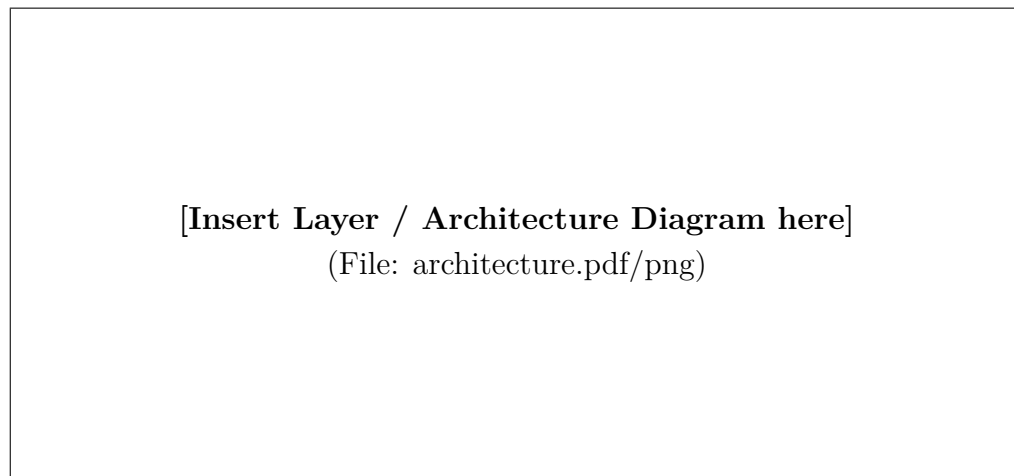


Figure 4.8: Layer / Architecture Diagram: System Structure Overview .

Chapter 5

Methodological Notes for Implementation

5.1 CV Parsing

Suggested approach:

- Preprocessing using regex.
- Heuristic post-processing to unify degree names, map skill synonyms.
- Optional: fine-tune Layout-aware models on a subset of academic CVs.

5.2 Matching and Ranking

- Use sentence / document embeddings for semantic similarity between job description and candidate profile.
- Combine similarity score with weighted heuristics: university ranking, years of experience, publication count, teaching experience, domain certifications.
- Train a learning-to-rank model (LambdaMART or pairwise ranker) if labeled match data becomes available.

5.3 Automated Testing

- Use existing question bank. Generate variations via LLM prompt templates.
- For objective questions: auto-grade.
- Set threshold (e.g., 40%) for progression to interview stage (configurable).

5.4 AI Interviewer

- Design interview flows focusing on behavioral prompts and situational questions.

- Capture audio and transcripts; apply multimodal analysis for confidence (prosodic features) and text-based personality signals.
- Always record interviews and allow HR to review and override AI assessments.

5.5 Fairness and Explainability

- Log model decisions and feature contributions for each candidate (SHAP or local explanations).
- Implement fairness audits (statistical parity, disparate impact) on historical data.
- Provide configurable attribute weights to allow policy-driven adjustments.

Chapter 6

Iteration Plan

This chapter outlines the planned development approach for the SAGE (Smart Academic Hiring System) project. The work is divided into four major checkpoints across FYP-1 and FYP-2. Each phase has specific goals related to system analysis, design, implementation, and evaluation.

6.1 Midterm FYP 1

During this phase, the focus is on understanding the academic hiring domain, studying existing AI recruitment systems, collecting requirements from HR stakeholders, and completing the literature review. Initial system design artifacts—such as use case diagrams, system flow, and high-level architecture—are also produced to build a foundational understanding of SAGE.

6.2 Final FYP 1

This phase includes the finalization of the Software Requirements Specification (SRS) and the development of core system components such as the CV parsing module, candidate-job profiling structure, and early matching prototype. The UI layout for HR and candidates is also drafted, and basic backend endpoints are implemented.

6.3 Midterm FYP 2

At this stage, major system functionality is developed. This includes implementing the improved Gale-Shapley matching engine, integrating LLM-based screening test generation, building the AI interview module, and connecting parsed CV data with ranking logic. Initial internal testing of all modules is conducted to validate correctness.

6.4 Final FYP 2

The final phase includes polishing the system, completing integration of all modules, performing system-wide testing, preparing the user manual, and deploying the SAGE platform for demonstration. Performance optimization and security verification in the backend are also completed.

Chapter 7

Iteration 1

The first iteration of SAGE focuses on system understanding, structural design, and the early development of core modules responsible for academic data handling and candidate-job matching. The emphasis is on representing key entities, modeling workflows, and designing foundational backend logic.

7.1 Domain Model/Class Diagram

The domain model for SAGE captures the essential entities used in academic hiring. These include Candidate, ParsedCV, JobPosting, Application, TestRecord, and MatchProfile. Each entity stores critical information required to evaluate a candidate, match them with suitable positions, and support HR decision-making. The model also highlights relationships such as one candidate submitting multiple applications and one job posting receiving many applicants. This conceptual foundation guides backend implementation and database schema design.

7.2 Component Diagram

The component diagram for this iteration outlines the major software components of SAGE:

- **Authentication Component** – Handles secure login and registration for HR and candidates.
- **CV Parsing Component** – Extracts structured academic data such as degrees, skills, publications, and experience.
- **Job Posting Component** – Enables HR to define job criteria and academic requirements.
- **Matching Engine Component** – Computes initial compatibility scores between applications and job postings.
- **Application Management Component** – Handles candidate submissions and

tracks application progress.

These components communicate via REST APIs, ensuring modularity and future expandability.

7.3 Layer Diagram

The system architecture is divided into three major layers:

- **Presentation Layer:** Web interfaces for candidates and HR to interact with SAGE.
- **Application Layer:** Implements business logic including parsing, scoring, matching, and validation.
- **Data Layer:** MongoDB-based storage for structured candidate data, job postings, parsed CV elements, and test performance.

This layered structure ensures maintainability and supports scalable deployment.

7.4 Structure Chart

The structure chart breaks down SAGE's main functions into hierarchical modules. The top-level HiringProcess module branches into UserManagement, CVProcessing, JobPostingManagement, and ScoringAndMatching. Each sub-module is responsible for specific tasks such as validating CV formats, extracting academic attributes, creating job profiles, and computing compatibility scores.

Behavior Design

7.5 Flow Diagram

The flow diagram describes the movement of candidate interactions: registration → CV submission → parsing → job selection → application submission. It also reflects HR workflows, which include job creation, reviewing parsed CV data, and observing initial candidate rankings. This structured flow ensures an efficient and transparent hiring pipeline.

7.6 Data Flow Diagram (DFD)

The DFD illustrates how data travels through SAGE. Candidate inputs flow to the parsing module, which outputs structured academic fields to the database. HR inputs

job requirements, which flow to the job-posting module. The matching engine combines parsed CV profiles and job requirements to generate ranking results for HR.

7.7 Data Dictionary

Iteration 1 defines core data elements such as:

- **CandidateID** – Unique identifier for each candidate.
- **JobID** – Identifier for each academic job posting.
- **SkillsExtracted** – A structured list of skills from the CV.
- **EducationLevel** – Extracted degree and institution information.
- **InitialMatchScore** – Early-stage score before interview and testing.

These definitions maintain consistency across modules.

7.8 Activity Diagram

The activity diagram outlines the candidate workflow from account creation to submitting a job application. It captures decision points such as CV parsing success, qualification validation, and job eligibility checks. HR-based activity flows include posting new positions and reviewing applicant scores.

7.9 Network Automata/Graphs or State Machine

The state machine represents the lifecycle of a job application: Draft → Submitted → Parsed → Scored → Under Review. State transitions occur based on parsing completion, job application submission, or HR evaluation events. This ensures accurate tracking of application progress.

7.10 Call Graph or Sequence Diagram

The sequence diagram for Iteration 1 focuses on the CV upload workflow. The steps include candidate upload → backend receives file → parsing engine extracts data → results stored in the database → confirmation returned to candidate. This clarifies the order of module interactions.

7.11 Interaction Overview Diagram

This diagram summarizes candidate registration, CV parsing, job posting creation, and initial scoring into one integrated flow. It shows simultaneous processes such as

multiple candidates submitting CVs and HR posting roles.

7.12 Schema Design/ER Diagram

The ER diagram defines the primary tables—Candidates, ParsedCV, JobPostings, Applications, and TestRecords. Key relationships include many-to-one (many applications to one job posting) and one-to-one (one candidate to one parsed CV). These relationships support efficient query execution in the hiring workflow.

7.13 Data Structure Design

Data structures in Iteration 1 focus on JSON-based schemas for storing academic attributes. Arrays represent publications and skills, while nested objects store education history. These structures simplify ranking and allow flexible parsing from diverse CV formats.

7.14 Algorithm Design

The initial matching algorithm computes a preliminary score based on skill overlap, degree match, and academic experience. Although simple, this scoring method prepares the system for integrating the improved Gale–Shapley stable-matching algorithm in later iterations.

7.15 Development Phase

Development tasks in Iteration 1 include setting up backend frameworks, implementing authentication, configuring the CV parsing module, defining API routes, and establishing naming conventions. Static analysis tools ensure clean code formatting and maintainability.

7.15.1 Unit Test

Unit tests validate functions such as registration validation, file upload handling, and parsing output consistency. These tests ensure that early-stage bugs are captured before integration.

7.15.2 Suites or Test Cases

Test cases include:

- TC01 – Successful account creation

- TC02 – Invalid CV upload format
- TC03 – Correct extraction of academic skills and education

7.16 Maintainable Phase

7.16.1 CI/CD

A CI pipeline is configured using GitHub Actions to run tests and style checks automatically. This ensures stable and consistent deployment of updated code.

7.16.2 Deployment Diagram

The backend is deployed on a cloud server (AWS EC2 or similar) with MongoDB Atlas as the database layer. The frontend is accessible through a web interface. This setup supports scalability and secure handling of candidate data.

7.16.3 System-Level Test Suites, Test Cases

System-level tests check the full workflow: candidate registration → CV parsing → job application → scoring. These tests validate the system as a whole.

7.16.4 SVN or GitHub (Optional)

The SAGE project repository is maintained on GitHub for version control, collaboration, and issue tracking.

7.16.5 Configuration/Setup and Tool Manual (Optional)

Local development requires Node.js, MongoDB, and necessary packages. Developers initialize the system using:

```
npm install
npm run dev
```

Chapter 8

Conclusion

This review shows that while components for automated hiring exist, an integrated, auditable system tailored to academic hiring (with CV heterogeneity, publications, and university-quality signals) is still missing. Sage plans to combine best practices from parsing, semantic matching, LLM-based assessment, and multimodal interview analysis while maintaining human oversight and fairness controls.

References

- [1] Elviwani, Siahaan, A. P. U., & Fitriana, L. (n.d.). *Performance-based Stable Matching using Gale–Shapley Algorithm*.
- [2] Suakanto, S., Siswanto, J., Prasetyo, I. R., Hardiyanti, M. (2021). *Interview Bot for Improving Human Resource Management*. In 2021 Int. Conf. on ICT for Smart Society (ICISS), pp. 4644–4647. IEEE. DOI: 10.1109/ICISS53185.2021.9533248.
- [3] Mendoza, A. A. D., Barbosa, L. J. C., Cortez, D. M. A., Dioses, R. M., Agustin, V. A., Regala, R. C. (2022). *Enhancement of Gale–Shapley algorithm with imbalanced sets for hiring and job finding applications*. Indonesian Journal of Electrical Engineering and Computer Science, 27(2), 954–958. DOI: 10.11591/ijeecs.v27.i2.pp954-958.
- [4] Will, P., Krpan, D., & Lordan, G. (2023). *People versus Machines: Introducing the HIRE Framework*. Artificial Intelligence Review, 56, 1071–1100. Accepted: 13 April 2022. DOI: 10.1007/s10462-022-10193-6.
- [5] Thangaramya, K., Roselind, J. D., Logeswari, G., Ahirwar, N., & Gajendran, S. (2024). *Automated Resume Parsing and Ranking using Natural Language Processing*. In 2024 3rd IEEE Int. Conf. on Artificial Intelligence for Internet of Things (AIIoT). DOI: 10.1109/AIIoT58432.2024.10574696.
- [6] Hosseini, H., Roy, S., & Zhang, D. (2024). *Putting Gale & Shapley to Work: Guaranteeing Stability Through Learning*. In 38th Conf. on Neural Information Processing Systems (NeurIPS 2024).
- [7] Aly, A. M., El Alem, A., Alawida, M., AlBloushi, S., & Aldhaen, G. (2025). *Optimizing Recruitment with AI: A Smarter Approach to Candidate Evaluation*. In

2025 1st Int. Conf. on Computational Intelligence Approaches and Applications (ICCIAA). DOI: 10.1109/ICCIAA65327.2025.11013684.

- [8] Mujtaba, D. F., & Mahapatra, N. R. (2025). *Fairness in AI-Driven Recruitment: Challenges, Metrics, Methods, and Future Directions*. arXiv preprint arXiv:2405.19699v3.

Chapter A

Appendix A: Example prompts and rubric

A.1 LLM prompt templates

A.2 Scoring rubric